

ARNAV CHAUHAN

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CAREER OBJECTIVE

Motivated Computer Science student with a strong foundation in Data Structures, Algorithms, and Full-Stack Development. Eager to leverage technical expertise and problem-solving skills to contribute to innovative projects and drive growth within a dynamic organization.

TECHNICAL COMPETENCIES

- **Languages:** C, C++, SQL, HTML, CSS, JavaScript
- **Frameworks & Libraries:** FTXUI, React 19, Next.js, Node.js, Express.js, PostgreSQL, Socket.io, Tailwind CSS, Framer Motion
- **Tools & Concepts:** Data Structures and Algorithms (DSA), Multithreaded Systems Architecture, CMake, Vercel, Render, Git

KEY PROJECTS & EXPERIENCE

Personal Portfolio Website

2026

Interactive Developer Portfolio / Full-Stack Developer

- Built a dual-mode developer portfolio in Next.js (App Router) and Tailwind CSS a standard scannable view plus a fully simulated desktop OS (window manager, taskbar, draggable windows) animated with Framer Motion.
- Integrated live GitHub API data (contribution graph, repo stats) and structured Problem / Role / Technical-Decisions write-ups per project, sourced directly from each project's own codebase; deployed on Vercel.

C++20 Interactive Developer Shell

May 2026

Systems Developer

- Architected a TUI terminal emulator in C++20/FTXUI around a Command-pattern registry (unique_ptr-owned, hash-mapped dispatch) and a hand-written, quote/escape-aware tokenizer built without regex.
- Ran external commands on a background thread with a lock-free, move-based handoff to the UI thread - batching output every 100 lines/100ms and capping scrollbar at 500 lines so the shell stays responsive under heavy output.
- Added Tab completion (commands and filesystem paths via std::filesystem), arrow-key command history, and a CMake FetchContent build with zero vendored dependencies.

Chess

2026

Full-Stack Multiplayer Chess Platform / Full-Stack Developer

- Built a real-time multiplayer chess platform (PostgreSQL, Express, React, Node.js) with Socket.io game synchronization, server-side move validation via chess.js, and Stockfish (WASM) AI opponents across difficulty levels.
- Implemented multi-provider OAuth (Google & GitHub via Passport.js) and validated every socket and API payload with Zod, since a typed frontend alone can't guard a live socket connection from malformed or malicious input.
- Split-deployed the frontend (Vercel) and backend (Render), since persistent Socket.io connections and session-based auth are incompatible with serverless functions.

PROBLEM SOLVING & DSA PRACTICE

- **LeetCode:** Solved 450+ problems focusing on Dynamic Programming and Graph Data Structures.
- **GeeksforGeeks:** Solved 150+ problems.
- **HackerRank:** Actively solving algorithmic challenges and participating in problem sets.
- **Competitive Programming:** Regular participant on Codeforces and AtCoder.

EDUCATION

Chandigarh University, Gharuan

2023 – 2027 (Ongoing)

B.E. Computer Science and Engineering | CGPA: 7.82/10 (till 6th sem)

D.A.V Centenary Public School, Haridwar

Intermediate (CBSE): 77.8% | Matriculation (CBSE): 74.6%

ACHIEVEMENTS & EXTRACURRICULARS

- **Team Leadership:** Reached Round 2 of the Tech Invent PPT competition as team lead.
- **Innovation:** Showcased a rotating solar panel project at a University-level technical exhibition.
- **Certifications:** Microsoft Azure Fundamentals, Design Thinking & Creativity for Innovation.
- **Engagement:** Organized and participated in university gaming, yoga, and wellness events.